MATH3018 Assignment 7, due on Apr 16

Questions (1)-(3) use the same dataset. The correlation matrix of a $p = 6$ population is

$$\begin{pmatrix} 1 & & & & & \\ 0.505 & 1 & & & & \\ 0.569 & 0.422 & 1 & & & \\ 0.602 & 0.467 & 0.926 & 1 & & \\ 0.621 & 0.482 & 0.877 & 0.874 & 1 & \\ 0.603 & 0.450 & 0.878 & 0.894 & 0.937 & 1 \end{pmatrix}.$$

The following estimated factor loadings are obtained by the maximum likelihood method:

	Estimated Factor Loadings $\hat{l}_{jk}$			
Standardized	Unrotated		Varimax Rotated	
Variables	F_1	F_2	F_1^*	F_2^*
Z_1	0.602	0.200	0.484	0.411
Z_2	0.467	0.154	0.375	0.319
Z_3	0.926	0.143	0.603	0.717
Z_4	1.000	0.000	0.519	0.855
Z_5	0.874	0.476	0.861	0.499
Z_6	0.894	0.327	0.744	0.594

(1) (4pts) This question uses the unrotated estimated factor loadings. Find the maximum likelihood estimates of the followings:
(a) The communalities.
(b) The specific variances.
(c) The proportion of variance explained by each factor.

[Solution]

(a) $\hat{h}_j^2 = \hat{l}_{j1}^2 + \hat{l}_{j2}^2, j = 1, 2, \ldots, 6$. Thus we have $\hat{h}_1^2 = 0.402, \hat{h}_2^2 = 0.242, \hat{h}_3^2 = 0.878, \hat{h}_4^2 = 1, \hat{h}_5^2 = 0.990, \hat{h}_6^2 = 0.906$.

(b) $\hat{\psi}_j = 1 - \hat{h}_j^2, j = 1, 2, \ldots, 6$. Thus we have $\hat{\Psi} = \text{diag}(0.598, 0.758, 0.122, 0, 0.010, 0.094)$.

(c) Proportion of variance explained by the 1st factor is

$$\frac{\sum_{j=1}^{6} \hat{l}_{j1}^2}{6} = \frac{4.001}{6} = 0.667.$$

Proportion of variance explained by the 2nd factor is

$$\frac{\sum_{j=1}^{6} \hat{l}_{j2}^2}{6} = \frac{0.418}{6} = 0.070.$$

(2) (4pts) Is the proportion of variance explained by the factor F_1^* the same as that of F_1? How does the total proportion of variance explained by F_1^*, F_2^* compared to that of F_1, F_2? Provide some justifications for your conclusions.

[Solution] The proportion of variance explained by the factor F_1^* can be different with that of F_1. Denote the i-th row of the estimated loading $\hat{L}$ by $\hat{\ell}_i = (\hat{\ell}_{i1}, \hat{\ell}_{i2})^\top$, and $e_1 = (1, 0)^\top$, then the variance explained by unrotated factor F_1 is

$$\sum_{i=1}^{6} \hat{\ell}_{i1}^2 = e_1^\top \left(\sum_{i=1}^{6} \hat{l}_i \hat{l}_i^\top \right) e_1.$$

After rotation by an orthogonal matrix Q, the new loading matrix is $\tilde{L} = \hat{L}Q$. Let q_1 be the first column of Q. The variance explained by the rotated factor F_1^* is

$$\sum_{i=1}^{6} (\hat{\ell}_i^\top q_1)^2 = q_1^\top \left(\sum_{i=1}^{6} \hat{l}_i \hat{l}_i^\top \right) q_1.$$

The above two quantities are generally different, thus the proportion of variance explained by the factor F_1^* can be different with that of F_1.

The total proportion of variance explained by F_1^*, F_2^* is the same as that of F_1, F_2. Because the total variance equals $\mathrm{tr}\hat{L}\hat{L}^\top = \mathrm{tr}\hat{L}QQ^\top\hat{L}^\top = \mathrm{tr}\tilde{L}\tilde{L}^\top$.

(3) (3pts) Using the principal component method we also get a model with two factors F_1^{**} and F_2^{**}. Is the proportion of variance explained by the factor F_1^{**} the same as that of F_1? Which one is larger? How does the total proportion of variance explained by F_1^{**}, F_2^{**} compared to that of F_1, F_2? Provide some justifications for your conclusions.

[Solution] No, usually they are different. The variance explained by F_1^{**} is larger because principal component method is to maximize the variance explained by each factor. Also due to this, the total proportion of variance explained by F_1^{**}, F_2^{**} is typically greater than that of F_1, F_2.

(4) (4pts) Establish the inequality

$$\left\|\mathbf{S}-\left(\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}+\hat{\boldsymbol{\Psi}}\right)\right\|_{HS}^{2} \leq \sum_{i=m+1}^{p} \hat{\lambda}_i^2,$$

where $\|\mathbf{A}\|_{HS} = \left[\operatorname{tr}\left(\mathbf{A}\mathbf{A}^{\top}\right)\right]^{1/2}$ is the Hilbert-Schmidt norm. Here $\hat{L}$ and $\hat{\Psi}$ are the principal component estimators, $\hat{\lambda}_i$ is the i-th largest eigenvalue of $\mathbf{S}$. (See page 17 of Chapter 7 slides)

Hint: Since $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}-\hat{\boldsymbol{\Psi}}$ has zeros on the diagonal, (sum of squared entries of $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}-\hat{\boldsymbol{\Psi}}$) $\leq$ (sum of squared entries of $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}$)

[Solution]

Since $\hat{\boldsymbol{\Psi}}$ is the diagonal part of $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}$, we have (sum of squared entries of $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}-\hat{\boldsymbol{\Psi}}$) $\leq$ (sum of squared entries of $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}$), i.e., $\|\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}-\hat{\boldsymbol{\Psi}})\|_{HS}^2 \leq \|\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}\|_{HS}^2$

The eigendecomposition of $\mathbf{S}$ is given as $\hat{U}\hat{\Lambda}\hat{U}^{\top}$, where $\hat{\Lambda} = diag(\hat{\lambda}_1, \cdots \hat{\lambda}_p)$ and $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots, \geq \hat{\lambda}_p$. Let $\hat{\Lambda}_1 \in R^{m\times m}$ be the upper left block of $\hat{\Lambda}$ consisting of the first m largest eigenvalues and $\hat{\Lambda}_2 \in R^{(p-m)\times(p-m)}$ be the bottom right block. Correspondingly, let $U_{(1)} \in R^{p\times m}$ consist of the first m eigenvectors $(\hat{u}_1, \cdots, \hat{u}_m)$ and $U_{(2)} \in R^{p\times(p-m)}$ consist of the remaining eigenvectors. Then $\hat{\mathbf{L}}$ can be written as $U_{(1)}\hat{\Lambda}_{(1)}^{1/2}$.

We have $\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top} = \hat{U}\hat{\Lambda}\hat{U}^{\top} - \hat{U}_{(1)}\hat{\Lambda}_{(1)}\hat{U}_{(1)}^{\top} = \hat{U}_{(2)}\hat{\Lambda}_{(2)}\hat{U}_{(2)}^{\top}$ which has sum of squared entries:

$$\begin{aligned}
\left\|\mathbf{S}-\hat{\mathbf{L}}\hat{\mathbf{L}}^{\top}\right\|_{HS}^{2} &= \operatorname{tr}\left[\left(\hat{U}_{(2)}\hat{\Lambda}_{(2)}\hat{U}_{(2)}^{\top}\right)\left(\hat{U}_{(2)}\hat{\Lambda}_{(2)}\hat{U}_{T(2)}^{\top}\right)^{\top}\right] \\
&= \operatorname{tr}\left[\hat{U}_{(2)}\hat{\Lambda}_{(2)}\hat{U}_{(2)}^{\top}\hat{U}_{(2)}\hat{\Lambda}_{(2)}^{\top}\hat{U}_{(2)}^{\top}\right] \\
&= \operatorname{tr}\left[\hat{U}_{(2)}\hat{\Lambda}_{(2)}\hat{\Lambda}_{(2)}^{\top}\hat{U}_{(2)}^{\top}\right] \\
&= \operatorname{tr}\left[\hat{\Lambda}_{(2)}\hat{\Lambda}_{(2)}^{\top}\hat{U}_{(2)}^{\top}\hat{U}_{(2)}\right] = \operatorname{tr}\left[\hat{\Lambda}_{(2)}\hat{\Lambda}_{(2)}^{\top}\right] = \hat{\lambda}_{m+1}^2 + \hat{\lambda}_{m+2}^2 + \cdots + \hat{\lambda}_p^2
\end{aligned}$$

Therefore we conclude the proof.